CircuitLab: A System for Aiding Remote Electronics Labs

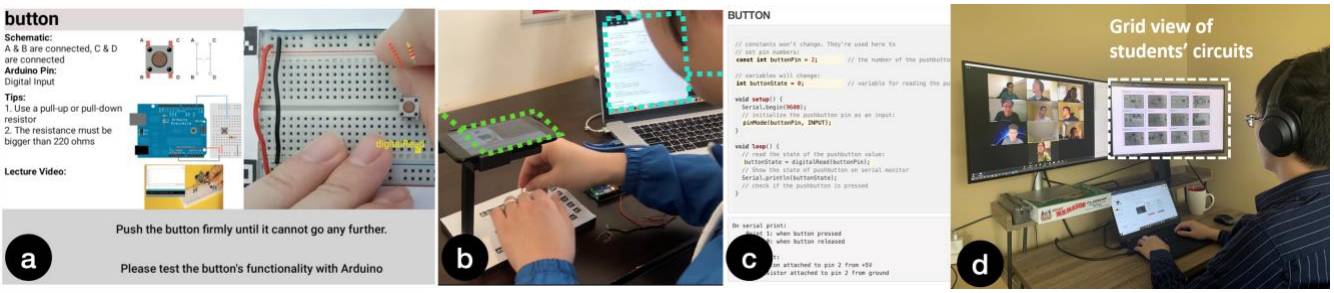

Figure 1. An overview of the CircuitLab system. (a) A smartphone is used for sensing circuit components and providing the students with context-sensitive learning assistance without monopolizing an instructor's time; (b) A students view of the CircuitLab setup, with a smartphone using a custom stand, and a laptop; (c) The laptop screen is used to provide the students with sample Arduino code for an inserted component; (d) A webapp interface for the remote instructor to interact and monitor students' progresses.

ABSTRACT

As remote learning scenarios have quickly become necessary, participating and learning in remote exercises (or labs) for circuit prototyping using video conferencing is still a challenge for students, as a remote instructor may not be able to manage a class effectively and efficiently when multiple students need assistance. To mitigate this challenge, we propose to augment existing video conferencing tools with a system continuously provides remote students with context-sensitive learning assistance (e.g., best practices with a component) as they are constructing a circuit. As part of the system, an instructor can also control whether learning assistance should be presented ambiently or through notifications. This allows the class to be better managed and students to be better engaged. To evaluate the effectiveness of our system, we ran a deployment study with an instructor mentoring nine remote students that were constructing a breadboard circuit. We report the results of this study and conclude by discussing the lessons learned from this work, as well as insights and future research directions.

Author Keywords

Breadboards, Circuit Prototyping, Mixed Reality, Remote Learning, Education Tools

CSS Concepts

• Human-centered computing~ User Studies, Interactive systems and tools

INTRODUCTION

With the recent mass proliferation of low-cost, and easily accessible electronic components and micro-controllers (e.g. Arduino [34]), novices with non-electronics backgrounds (e.g., students in computer science, design, or arts) have been empowered to learn electronics prototyping and circuit design through classes offered by colleges or maker communities. A critical component of students' learning activities often involves in-class exercises (or labs), which allow students to practice their skills and better understand the concepts taught in classes. Although many existing labs (and teaching materials) are designed to be delivered in-person, there is now an increasing need for the students to join remotely for improved digital access in education and in situations where face-to-face interactions need to be avoided entirely (e.g., during a pandemic).

With current technologies, remote labs can be held with students and an instructor connected through video conferencing tools (e.g., Zoom [42]). Each student constructs their circuit locally, which can then be shown to the instructor using the camera on their laptop or a smartphone for feedback or advice on debugging. Students can also share their code using screen sharing. However, the issue is that many of the existing video conferencing tools are not designed for remote electronics labs. It is thus unclear how effective they are for students to learn and practice their skills in a remote setting.

To understand the challenges for both students and instructor, we conducted a formative study with nine students, each of whom created a breadboard circuit in their homes, while an instructor was made available remotely to answer questions and provide advice. Through the study, we observed that a significant amount of time and effort was spent by the instructor to answer simple or continuously repeated questions, understand students' problem, and provide help. This meant that many students did not have an opportunity to interact with the instructor and receive help useful for them to complete the lab exercise and thus their desired training.

To mitigate these issues, we designed and developed CircuitLab, a tool that builds on top of the concept of existing video conferencing tools to better facilitate remote

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electronics labs. With our system, students and an instructor are connected through a video conferencing tool using their laptop and a smartphone (that is held 13 cm above a breadboard) for sharing the circuit and sensing the circuit components using computer vision (Figure 1b). This allows the system to act as a virtual instructor, providing students with context-sensitive learning assistance (e.g., reminder of the operating principle of a component or suggested circuit styles), as well as become scalable to the entire class without monopolizing an instructor's time. When the system is in use, circuit-related learning resources are shown on the smartphone and programming related resources (Arduino code for the inserted components) are shown on the laptop (Figure 1c). An instructor can view the students' circuits (Figure 1d). Before the class, the instructor can configure whether learning assistance should be presented ambiently or through notifications. This allows the class to be better managed and students to be better engaged.

To demonstrate the effectiveness of our system, we conducted a deployment evaluation with nine students and an instructor in a second remote lab scenario, through which, we found that our system could mitigate many of the issues identified from the formative study. The students found our system useful in getting the answers for common questions at their own pace. This allowed students to spend more time on designing, practicing, and learning the circuit of the lab. For the instructor, we found that lab time could be better managed. More time could be spent for the instructor to proactively seek engagement with more students.

The main contributions of this paper are: 1) insights into the current challenges of practicing electronics and circuits in a remote lab context; 2) a novel system that facilitates remote electronics labs; and 3) lessons from a deployment study investigating initial impressions of the system's effectiveness from the instructor and student perspective.

RELATED WORK

Our work is informed by and extends prior research in circuit prototyping and debugging tools, remote learning, and ambient system design.

Tools for Circuit Prototyping

From mis-wiring and bad seating, to missing or incorrect components, novices face several common errors and challenges when first starting to learn about circuits, electronics, and prototyping [5, 30]. Prior research has shown that circuit prototyping is hard to teach and learn by both instructor and student alike [11]. For some novices, these challenges are due to a mismatch in the actual functionality of the device with their own mental model of the device [15], or even confusion as to whether their error is hardware or software based [5].

This has given rise to a number of different software and hardware tools to support users with learning and debugging electronics and circuits. On the software side, tools such as Fritzing [21] , and others [2, 3] emphasize using a virtual breadboard as an interface for introducing novices into electronics. Several of these software tools allow users to test and debug circuit designs through simulation. The challenge with software-based tools, however, is that novices cannot physically interact with components, which is a limitation when first learning circuits and electronics[39] . Additionally, simulation tools do not easily handle complex real-time sensor input when testing functionality [19].

Alternatively, physical interaction is a key component of many of the hardware-based solutions in the research space, with a common approach focusing on using an augmented breadboard [13, 19, 38, 40, 41]. One of the key benefits of physical interaction with electronics is the notion of learning embodied skills that occur when students are learning with actual hardware [10, 22]. Tools such as Toastboard [13] help users sense and visualize voltage readings, while CurrentViz [40] allows a user to visualize the flow of current in a circuit. It is also even possible to detect what components a user has inserted into a breadboard [41]. While many of these tools have focused on making it easier for novices to overcoming some of the common errors that are encountered, many of the challenges novices face are due to a lack of experience in debugging [15, 30]. Work such as Heimdall [17] enables a remote debugging experience between a novice and an instructor, which has shown value in prior work [8]. The system, however, require ad-hoc hardware, sensors, and actuators, which may not be accessible to everyday users. Additionally, these tools thus far, have not been extended to remote classroom experiences with multiple students.

Learning and Management Systems

Common features of modern applications for remote learning include the ability for an instructor to monitor the screens of several students [35], and the ability to coordinate a large classroom [27, 36]. However, these types of tools do not necessarily offload the cognitive workload of an instructor, unlike CircuitStyle [11], which instead focuses on assisting an instructor with potential problems before they arise (related to circuit style errors) by providing contextual information. While it demonstrated value in reducing common errors of students and minimizing the workload for an instructor, it was not designed for remote scenarios.

Dashboards [14] and logs [24] are other techniques for instructors to manage a classroom effectively. For dashboards in the context of electronics (and especially remote electronics), it is important to allow an instructor to remotely monitor and support the interactions a student has with physical components, while also helping them to aid novices without their attention being monopolized [11]. Similarly, systems that use logs, such as CADament [25] and others [7, 31] improve the learning experience by providing additional help. What is under explored in this context, however, is how this approach can be extended to remote learning scenarios, especially involving electronics. Building upon [11, 17], we focus on extending these types of

approaches by augmenting video conferencing tools with continuous and context-specific learning assistance.

Ambient Systems

Ambient systems (and displays) are designed to limit the distraction of users by providing the ability to monitor noncritical information [29, 37]. A canonical example of an ambient system is Portholes, which provides users with an awareness of who is present for communication [12]. In designing systems such as these, information capacity, notification level, representational fidelity and aesthetic emphasis are key dimensions for consideration [32].

In the context of educational and learning resources, an ambient system should provide high representational fidelity, and limit the interruption of a user, similar to Ambient Help [28]. A major challenge with these systems, however, is the potential for information overload [18]. In remote learning scenarios involving electronics, an optimal amount of information needs to be provided for both the novice and the instructor. For the novice, the information should transition between both tasks that involve intense work (e.g. wiring components) or information processing (e.g. determining specifications of an electronics component). Buxton describes this transition in a framework of foreground and background activities [6]. Foreground activities such as listening to music on a mobile device are in the "fore of human consciousness", while being aware of someone listening to music nearby is a background activity, taking place "in the periphery 'behind' those in the foreground".

We build upon the principles of ambient system design and work such as Ambient Help [28] with the aim of providing ambient information for novices around circuit design, as well as for instructors who can easily monitor students without information overload. Next, we describe the formative study that shaped our system.

FORMATIVE STUDY: IDENTIFYING CHALLENGES

Building upon the literature, we first aimed to understand the challenges that students and instructors face in a remote lab where electronics prototyping is taught.

Participants

We recruited nine university students (2 female, 7 male, average age: 24.6) to remotely participate in our study. Five participants majored in Computer Science while the others majored in Applied Arts. Six had recent experience with simple circuit prototyping from makerspace workshops, while the others learned basic electronics in High School.

Procedure

We held a remote lab session with one instructor that has an electronics background and one-year of experience teaching circuit prototyping workshops to non-electrical engineering students at a University. We chose a circuit from the Arduino Project Starter Kit [20] for the exercise to be used in the remote lab session. The circuit was composed of a motor, a transistor, and a button. When powered, the circuit allows a user to use the button to turn the motor on or off, and to use the transistor to set the speed of the motor (Figure 2).

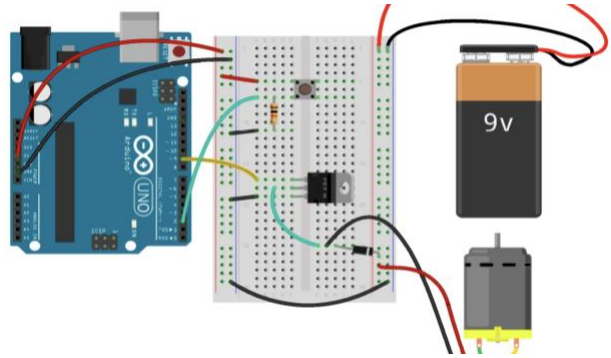

Figure 2. Target circuit for students in the formative study.

Prior to the lab session, all components were sent to students. As part of the lab session, instructors and students were required to be connected over video using Zoom through their laptops. The laptop cameras were also used by students to show the remote instructor their circuits. The first 20 minutes of the lab session was used by the instructor to go over the course materials, including basic electronic concepts (e.g., voltage divider), how to use a breadboard with Arduino, and the components to be used in the lab exercise (without wiring), and the goal of the exercise. Next, the students had 60 minutes to design and construct the desired circuit for the lab. Students were encouraged to ask the instructor for advice or questions on debugging their circuit or code. As the aim of this lab was for students to learn and practice, they were not formally graded or evaluated.

We recorded all conversations between the students and instructor, as well as their laptop screens for analysis. We also asked the students to log the challenges they had using the system. A 10-minute post-study interview was also conducted for the instructor and all students.

Results

In the hour that was provided, no students completed the circuit. Based on observations and interviews, we identified several major problems. Some are new to the remote setting while others were not, but their issues are more pronounced.

First, there was a significantly long wait time for students to receive help, as the instructors waitlist was long. The instructor also told us in the post-study interview that it took him longer to help students remotely. It was common for students to ask similar questions or on topics already covered in the class and handout, such as component specification (e.g., IC pin functions) or how to make a functional circuit for a component, as they were easy to miss. Some bugs were also common (e.g., miswiring a button). As such, time was spent by the instructor on either routine, simple, or repeated questions. This led to multiple students queuing up questions, and less students receiving help from the instructor. In our case, the instructor was only able to help 3 students for the entire period of the lab. There was a waitlist of 6 students

who were not helped when lab ended. We also foresee that with this issue, more experienced students could miss the opportunity to interact with the instructor on topics beyond the current exercise.

Second, it was significantly more challenging for the instructor to view and understand a student's circuit through a camera, especially when wires are messy. For example, when components or pin connections were occluded by wires, the instructor misinterpreted a student's circuit, which led to time being wasted by the instructor to search for errors in the wrong places. This ultimately impacted the students' learning experience. The instructor stated that practicing good circuit style behaviors could effectively mitigate this issue as breadboard circuits with good style are known to be more readable, less error-prone, and easier to debug [11]. Other good habits, such as constructing and testing I/O component circuit one at a time also lead to less errors. However, good behaviors were largely overlooked by the students. Some of the poor styles were corrected by the instructor but most were missed since the instructor was swamped with other questions.

Third, it was a common request from students to revisit and discuss some of their previously taught concepts that they believed to be useful for the current project, but time did not permit. For example, a student described in the post-study interview that he had learned how to wire a button previously, but still had forgotten to use a resistor. The student said that a reminder would have been very helpful in this situation for them to consolidate what they have previously learned outside the lab. Further, more time could have been spent on learning new materials rather than debugging an avoidable error. This was confirmed by the instructor, who also stated that students could have been more engaged in learning if he could proactively give them suggestions and advices on-the-fly. However, interacting with every student in the class was nearly impossible. The instructor suggested that a mechanism that could automatically help the students to avoid or debug some of the common errors could have largely mitigate these issues.

CIRCUITLAB

Based on the results of the formative study, we developed CircuitLab, a tool builds on top of the concept of the existing video conferencing tools to better facilitate remote electronic labs. To enable the use of commonly available devices, we developed our system using a smartphone app and a web app running on a laptop. With CircuitLab, students connect to the instructor via a video conferencing tool using their laptop and the smartphone – held by a 3D printed station 13 cm above a breadboard – is used for sharing a circuit (Figure 3). The station holds the smartphone and breadboard in place to ensure a stable view of the circuit, while also allowing the smartphone to sense circuit components (using computer vision), based on which, the system acts as a virtual instructor to provide students with context-sensitive assistance and information (e.g. reminder of the operating principle of a component, suggested circuit styles, etc.). In this way, the system is scalable to the entire needs of the class and assists the students in their own pace in practicing their electronic skills. Additionally, the instructor can determine the level of obtrusiveness for the information is presented to the students, allowing them to better manage a lab or class.

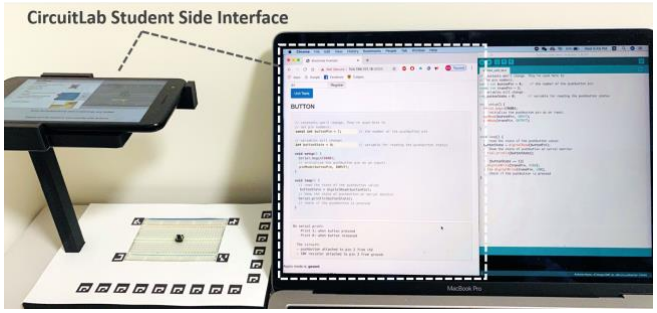

Figure 3. The CircuitLab student side includes a smartphone interface and a desktop interface.

CircuitLab Help Information

Our system provides four types of information, related to *component*, *module*, *circuit style*, and *unit test*.

Component Related Information.

Our system supports three most common types of components: (1) basic components, (2) I/O components, and (3) IC components. The basic components (e.g., resistors, capacitors, diodes, and transistors) are the primitive electronic components used as building blocks to form functional circuits (we call them modules) for the I/O and IC components. I/O components include input devices (e.g., buttons), sensors (e.g., pressure sensor), and output devices (e.g., speakers and motors). Finally, the IC components include varying types of integrated circuits (ICs), such as H-bridges and amplifiers. For each type of component, the system provides a student with information regarding its name, functionality, and how it works. For the I/O and IC components, the system also provides pin information. Showing component information can help in saving time answering some of the common or repetitive questions for an instructor.

Module Related Information

This type of information is primarily related to the functional module of a component, including how it can be constructed and its principle of operation. We do not show how different modules work together to form the target circuit because it is up to the students to figure out. Showing this information can reduce circuit errors and again save time on answering questions for an instructor.

Circuit Style Related Information

In our system, we also included a description and demonstration of a number of good circuit styles, such as "*avoiding crossed wires while prototyping*", "*using as little wire as possible*", "*checking the polarity of components before insertion*", and "*Keep pin 1 of the IC's pointing the same direction*". Good circuit styles lead to circuits that are

more readable, less error-prone, and easier to debug, which is arguably more important in a remote setting.

Unit Test Related Information

Unit tests encourage modularization and help avoiding errors. This category of information includes the module information discussed previously and Arduino code that is easily modifiable to test a constructed module. Through reading sample code, students can also acquire a better understanding of how the hardware and software components of a circuit work together.

CircuitLab Interface for Students

We show the assistance and help information on both the smartphone app and the desktop app. The smartphone app shows the information related to the breadboard circuit – sometimes with Augmented Reality (AR) – while the desktop app shows the information related to Arduino programing (e.g., sample code for a component on the breadboard) using a web app placed beside the student's Arduino IDE. The smartphone is physically held at a height where the screen is visible in the periphery of the student's vision when a circuit is being constructed. This enables a student to quickly find the information when needed by peeking at the screen or simply ignore it when their attention needs to be on the circuit.

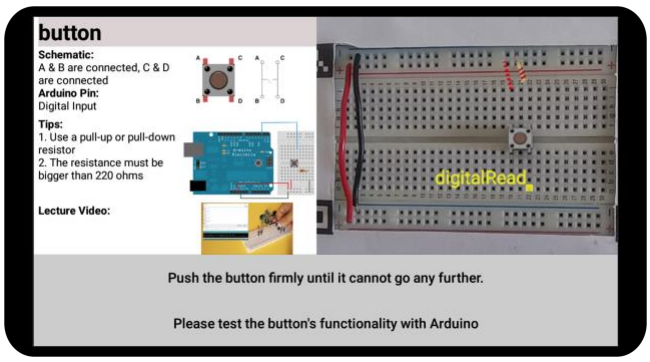

Figure 4. CircuitLab smartphone interface for students. Left panel shows component and module information in text, figure, and video. Right panel shows pin functions and module in AR. Bottom panel shows the suggestions for circuit style behaviors.

Assistance is provided based on the progress of each individual student and the components in their circuits. For example, when a student inserts a button into the breadboard, the smartphone app shows information regarding how a button works and, on the laptop screen, the code for unit testing the button (Figure 3 left). The smartphone app also shows the theory behind how a button works and a diagram demonstrating how it button can be wired to a resistor and an Arduino board to form a functional circuit (module, Figure 4). The student can also watch a demo video of how the module itself works (e.g., to control an LED, Figure 4). Also, using AR on the smartphone app, students can find information regarding the pin function of a button as well as how the button module can be constructed on their breadboard (Figure 4). A student can also swipe to view alternative ways a button module can be created.

Suggestions on good style behaviors are also provided progressively. For example, prior to the construction of the circuit, a student can be reminded to connect the power rails of the breadboard. After the button is inserted, the student can then be reminded to perform a unit test to check if the button works as expected. To do so, the student can simply modify the provided Arduino code (e.g., correct the pin numbers). Upon the student inserting another component into the breadboard, all the displayed information is updated to reflect the new component. If the student wants to revisit the information of a previously inserted component, the student can simply tap the component in the AR view.

CircuitLab Interface for Instructor

We also developed a web-based interface for the instructor to view the circuits of students and configure how and what information can be provided to the students (Figure 5). The instructor can view circuits as livestream videos or as images updated any time a change occurs in the circuits (e.g., adding or removing a component). The instructor can also choose the type of information that can be presented to the students, as well as how the student should be notified about it. For example, the instructor can enable students to be notified through a sound so that important information will not be missed. This can be helpful for consolidating previously learned knowledge or for reminding students about common errors (e.g., missing a resistor for an LED). Alternatively, the instructor can choose to let the information be presented ambiently. This way, the information is pushed to the students without interrupting them from work and can be viewed when needed.

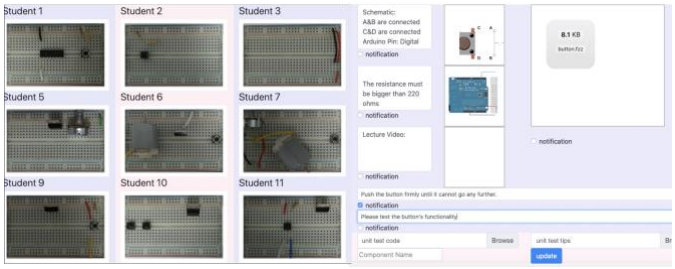

Figure 5. Web interface for instructor to monitor students' progress during class and manage information before class.

IMPLEMENTATION

The system is composed of three parts: (1) a smartphone app implemented on a LG Rebel 4, (2) a web app, and (3) a backend server. The smartphone camera captures the breadboard circuit. We left the Arduino board out from the camera view because the phone, optimized for ambient viewing, is not at a height that can capture a complete image of the Arduino board beside the breadboard. All the computation and machine learning are handled by a server running on Google Cloud. The web and smartphone apps are used for visualizing help and assistance. This allows the software to be lightweight on the client side.

Recognizing Electronic Components

We used MASK R-CNN [16] for recognizing and segmenting circuit components. We started with a model from Facebook AI Research, pre-trained on the COCO Dataset [26]. We then fine-tuned the model using our own dataset. We used the ResNet-50-FPN for the backbone of the Mask R-CNN model with the hyper-parameters remaining the same as the original setting in the paper [16]. The model was trained on Google Cloud with a Tesla K80 GPU for 12k iterations, starting from a learning rate of 0.02.

The training data covers the common components used by novices, which were identified by randomly sampling 500 circuits developed on a half-sized breadboard from Fritzing Projects [33], an open-source online community for sharing circuit projects. Among the 31 types of components found in these circuits, we chose the ones appearing 10 times or more for our implementation (19 in total, Figure 6). then recruited five volunteers to construct the circuits on a breadboard by following the Fritzing diagram. For the components that cannot be directly inserted into the breadboard (e.g., motor, speaker, piezo), we wired and placed them anywhere on or beside the breadboard. The top view image of the circuits was then captured for training using the volunteers' smartphones held by hand. We annotated the images using LabelMe [23]. Since this is the first known dataset for breadboard components, we made it publicly available[1]. Note that our implementation does not separate different models of the same component type. For example, our dataset has ICs with 8, 14, 16, and 28 pins and they are treated the same. It is fine for our application since it is rare for a lab circuit to have varying models of the same component.

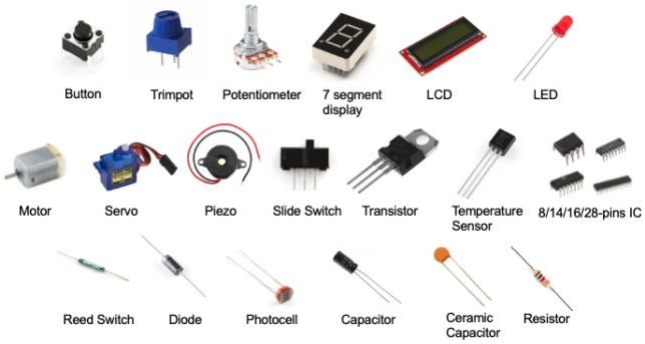

Figure 6. Nineteen test components used in our system.

Components recognition and segmentation is carried out only when there is a change in the breadboard circuit, such as insertion of a new component or removal of an existing one. To make it lightweight, the system infers there is a change in the circuit upon the occlusion of at least three of the ArUco markers placed around the breadboard (Figure 7), likely caused by the hand(s) interacting with the circuit. Once after the hands are clear from the camera view, the system captures an image of the breadboard to determine the new components or the ones that are removed. With the current implementation of our system using a single GPU server (GPU: Tesla K80, CPU: 2vCPU, RAM: 7.5G), component recognition can be carried out at 2 Hz per student's end for a class of twenty.

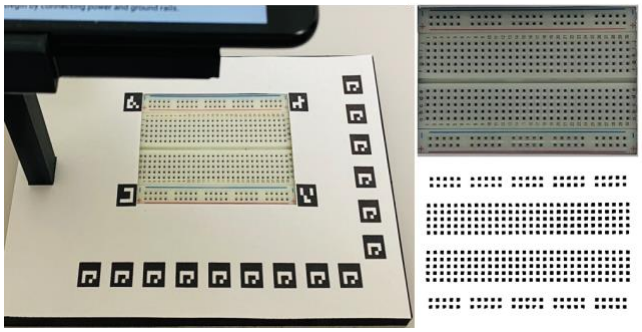

Figure 7. Left: The ArUco markers placed at the four corners of the breadboard are used to segment the breadboard from the camera view. The markers placed around the basis of the station are used for detecting hand occlusions. Right: Breadboard and the holes identified by our technique (bottom).

Determining Breadboard Hole Location

Showing help information using AR requires the virtual components and wires to be rendered at the right pins on the breadboard. Our system registers the location of the breadboard holes through an automatic calibration process on an empty breadboard upon the start of the system. It first segments the breadboard from the captured images using the ArUco markers [4] placed at the four corners of the breadboard (Figure 7). The pins are then identified based on the difference in the contrast between the holes and the white breadboard surface using an adaptive threshold method ([1]) in the HSV space (Figure 7).

Determining Component and Pin Location

In the AR view, some of the help information needs to be shown beside an I/O or IC component (e.g., pin functions). This requires the system to roughly estimate the location of the component and its pins. Component location is known through the mask predicted by the machine learning model. Estimating the location of the pins is challenging since the pins are small and are usually occluded by the component itself (Figure 8). However, I/O and IC component pins are often laid out in a fixed pattern, especially for those with short or stiff pins of three or more. For example, a transistor has three pins in a row, a button has four pins one at each corner. Since these components are connected to the breadboard through the holes right underneath them, our system estimates the location of their pins based on the detected breadboard holes inside the bounding box of the detected component. Among all the detected pins, we find a set, which follows the pattern of the pins of a recognized component. Within each set, the pins are equally spaced. If multiple sets are found, we randomly choose one. This method works for most types of the I/O and IC components supported by our system (9 out of 12 types) but it does not necessarily extend to the I/O components with leads (often two) that are long and bendable (e.g., LED, photocell, and

[1]CircuitLab dataset link: https://github.com/CircuitLab2020/CircuitLab_Dataset

reed switch) since the location of the connected pins are less predictable.

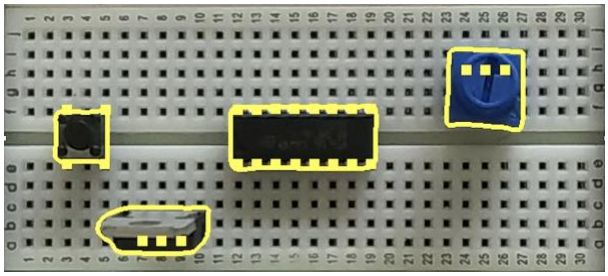

Figure 8. Illustration of component pins detected using our technique.

Component Data Collection

All the component information, pin functions, and suggestions on component modules are based on datasheet schematics of the components and the circuits used for training the machine learning model. The sample code and videos used are from tutorials found on the internet, but the instructor can also choose to use their own materials. A functional module of a component can be created in many different ways. The order of their presentation is based on how frequent they appeared in the collected circuits. The one ranked the first is shown in the system by default. Circuit styles are chosen from Davis, at al.'s work [11].

SYSTEM EVALUATION

We conducted a study to evaluate how accurately our system can recognize the selected components, locate breadboard holes and component pins.

Recognizing Components

We included in our study 19 types of components, as shown in Figure 6. The ICs with a different number of pins were considered the same type. Some types of components appeared more frequently than others in the collected circuits. For each type, we included roughly 80% of the samples for training with the remaining 20% for testing.

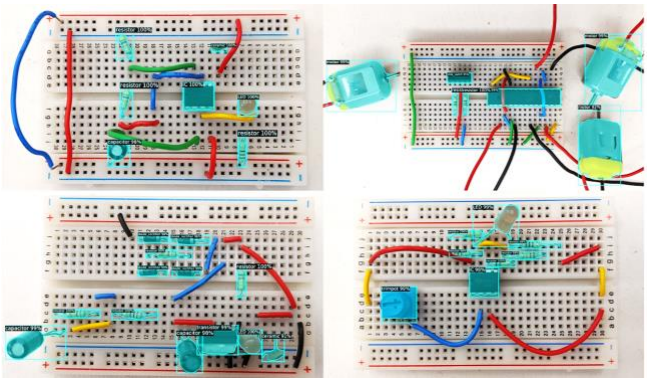

Figure 9. Sample images within detections and segmentations of component instances.

Result. As shown in Figure 9, our system was able to recognize and segment the tested components with good accuracy. We also report the results using average precision (AP), a numerical metric commonly used in object detection and segmentation. Table 1 shows the mask and bounding box APs. In comparison to the APs achievable on the COCO Dataset using the SOTA model (both around 50 [9]), our system achieved a higher value. Note that the two datasets are very different, so the APs are not meant to be compared directly to each other. We only show it as an indication of the promise of our system. Finally, there were no small objects in our dataset because all components were bigger than 32^2 pixels in the image.

AP	**AP_{50}**	**AP_{75}**	**AP_S**	**AP_M**	**AP_L**
72.3	97.5	82.9	N/A	43.1	72.2
AP_{bb}	**AP_{bb50}**	**AP_{bb75}**	**AP_{bbS}**	**AP_{bbM}**	**AP_{bbL}**
80.0	98.2	86.4	N/A	60.8	71.1

Table 1. The results of component recognition (second row) and segmentation (first row) using Mask R-CNN. AP_{50} and AP_{75} are the AP values calculated with an IoU threshold of 0.5 and 0.75 respectively. AP is the mean average precision calculated with IoU values ranging from 0.5 to 0.95 with an interval of 0.05. AP_S, AP_M, and AP_L are APs for small (area < 32^2), medium (32^2 < area < 96^2), and large (area > 96^2) object in the images.

Locating Breadboard Holes

We also tested how well the breadboard holes can be located by the system using our technique under different lighting conditions. Data was collected in a home environment illuminated by a ceiling light. The device was placed on a desk 2 meters away from the light in the horizontal distance. The light of the room was adjustable through a slide dimmer. We conducted the study under 6 different lighting conditions, ranging from 100 lux to 400 lux with an interval of 60 lux. The room was bright for work under the 300 lux condition and close to somewhat too dark for work in the 200 lux condition. The light intensity was measured beside the breadboard using Digital Luxmeter AS803.

Result. We counted the number of breadboard holes that are identified by the system, and we found that all the pins could be successfully detected under all the tested conditions.

Locating Component Pins

Finally, we tested how well the system can locate the pins of the nine I/O and IC components, as shown in Table 2, in their hosting circuits. Two volunteers were recruited to construct 56 circuits using our system by following the Fritzing diagram. Upon the completion of all the tested circuits, we manually labeled the pins of the tested components as ground truth, which was then used to compare with the estimated pin locations using our technique. The results are reported using Error Rate and Average Error Distance (Table 2). Error rate is the percentage of time an error occurred for a component. For example, if an error occurs on two of the three pins of a transistor, the error rate for the transistor is 50%. Since column holes are electronically connected, offsets along the column are not counted as an error. The average error distance measures the average distance between the estimated location and the ground truth. For example, if the

estimated location is two pins away for both pins, the average error distance of the transistor is 2.

Result. Among the tested components, the system was able to correctly identify all the pins of Side Switch and ICs without an error, despite that the ICs had the greatest number of pins with varying configurations (8, 14, 16, and 28). For the remaining components, our system achieved an error rate ranging from 9.1% to 50% (Table 2). Although some of the numbers look high (e.g., transistor and 7 Segment Display), the average error distance is in fact, low. This is because the estimated locations were equally spaced. If a pin was detected incorrectly, all the remaining ones are shifted by the same error distance. The results demonstrated that the average error distance was 1 for all the components with pin detection error, which indicates that the estimated pin locations for these components are one pin away from the ground truth. Therefore, users were informed about this potential error in the AR view.

Component (num. of pins)	Pin Instance	Error Rate	Avg. Error Distance
IC (with 8, 14, 16, 28 pins)	272 (80, 28, 80, 84)	0	0
Button (4)	100	16%	1
Temperature Sensor (3)	63	38.1%	1
LCD (16)	48	33.3%	1
Potentiometer (3)	33	9.1%	1
Transistor (3)	27	44.4%	1
7 Segment Display (10)	20	50%	1
Trimpot (3)	15	40%	1
Slide Switch (3)	12	0	0

Table 2. The result of the pin detection study reported in Pin Instance, Error Rate, and Average Error Distance.

WORKSHOP DEPLOYMENT

To measure the effectiveness of our system in a remote electronics lab, we conducted a deployment study. Our goal was to understand whether and how the system can facilitate learning and teaching experiences for students and lecturer.

Participants

We recruited nine computer science undergraduate students (2 female, 7 male, average age: 24) to participate in our study. All of them have learned basic electronics experience from high school. Four had prior experience with Arduino. None were involved in the prior study or the data collection.

Procedure

The lab was held remotely with the same instructor from the formative study. The students and instructor were connected through Google Meeting and our CircuitLab system. We used Google Meeting because Zoom was inaccessible to some of our participants due to the security policy of their home institute. The course material, circuit, handout, and study procedure remained the same as the formative study, but now students could spend as long as they wanted to complete the circuit. This allowed us to obtain deeper insight into how the CircuitLab system may or may not be able to assist them in a meaningful way. The instructor set all but the circuit style information to ambient mode, which means all the assistances were provided as ambient help except for the circuit style. Notifications for circuit style suggestions were sent to the students as a way to reinforce good behaviors.

Results

Overall, we found that the students enjoyed the lab through practicing the circuit using our system. The instructor also found it effective to mitigate several issues identified from the formative study. We discuss our key findings by first presenting the feedback from the students, followed by the feedback from the instructor.

Student Experience

Among the nine students who participated in the study, six were able to successfully complete the circuit within 48 minutes. This is better than the formative study, in which none of our participants was able to complete the same circuit within 60 minutes. The remaining three students spent roughly 75 minutes to complete the circuit.

The system helped the students with some of the common questions they had with the answers being available when needed, thus saving them time on searching. As such, more time could be spent on designing, practicing, and learning the circuit. When the students encountered questions, whose answer was not provided by the system, they turned to the instructor. Throughout the lab, five students sought help from the instructor. All questions were related to debugging. This is encouraging in the sense that many of the routine questions were well handled by our system, but on the flip side, remote debugging was still a time-consuming task. Students had to wait in a queue to get help from the instructor, which is a part of the reason why it took some students more than 60 minutes to complete the lab.

The time saved from the instructor addressing some of the common and repetitive questions allowed him to be proactive in seeking opportunity to give students suggestions and advice on how to better practice circuits and avoid errors. The students told us that this was also a part of the reason why many of them were able to complete the lab on time. More importantly, the student had the opportunity to discuss things with the instructor outside what was covered in the lab materials. Overall, the system scored 6.2 with respect to how useful it was as an addition to the video conference tool (1 being not useful at all and 7 being very useful; SD: 0.4). Next, we discuss the feedback from the students with respect to the four types of information provided by the system.

Component-related Information. The students found it helpful to see how a motor/button/transistor works. Many students used this information proactively. Eight of them mentioned that receiving such information on-the-fly was a useful mechanism for them to better understand the concepts taught in the class. One student described: *"I did not learn well on the concepts taught in the lecture until I started to*

build the circuit. With the assistances from the system, I could recall what were taught and connect them to the component that I was actually using."

For many students, showing pin function for the transistor was the most helpful feature in this category. They found it a convenient reference to use, which also helped them focus on the design of the circuit rather than small details. A student with previous experiences in Arduino prototyping mentioned: *"It was really convenient for the pin functions to be shown in AR, so I did not have to stop what I was doing and search in the handout."*

Module-related Information. Many students found this category of information helpful in learning how the components work together and reducing circuit errors. Six students said that they enjoyed this feature. Among them, three were novices, who had no previous experience in circuit prototyping. They deemed it as the most useful feature of the system, as it provided them with the confidence needed to try and experiment different circuit designs in their mind. One student described: *"I was not so sure about if it was the right way to wire the motor and was afraid that I might burn it. The system was handy in that it constantly showed what was the right thing to do."*

We were also interested in knowing if providing the students with wiring information would have defeated the purpose of learning since the students could have replicated the module without understanding how it works. Many of the students told us that it was not the case since they were motivated to learn. Together with the component information, students found it helpful for consolidating their understanding of the component and how it works (e.g., why the transistor pins should be wired this way). All students understood that the suggested completion of a component module may be incorrect in pin location in AR, but they did not find it a problem since the suggested component to complete the module was considered more useful than its absolute location.

Further, the suggestions on module completion was used by the students to check the correctness of their circuit and identify bugs. This was particularly true for the students, who had prior experience in Arduino, as some of them could develop part of the circuit based on previous knowledge. A student told us she was able to quickly find a bug in the button circuit by peeking at the suggested module even though their implementation was slightly different.

Circuit Style Related Information. This category of information was provided in a more obtrusive way as the students were notified about the styles they should practice as they progress through their circuit. We observed that at least four students followed and practiced most of the suggested styles. For example, the students ensured that jumpers were trimmed to lay flat against the breadboard and double checked if the components were inserted firmly. Amongst these students, two had previous Arduino experiences with one student describing to us: *"I knew some of the circuit styles before but seldomly use them as it was something easy to slip, especially when the functionality of my circuit was my top priority. I think this system is handy in reminding me of these styles."*

We think this is a promising start as reinforcing good circuit style behaviors is a known challenge [11] in teaching electronics, despite the fact that the benefits are well-aware. We found that all four students, who confirmed practicing circuit styles, spent an average of 29 minutes (SD = 7) to successfully complete the circuit. This is shorter than the class average of 41 minutes (SD = 19). It is important to note that the instructor was swamped with questions in the formative study, and thus unable to provide the students in need with detailed suggestions. Our system is scalable to the entire class, allowing everyone to be reminded.

Unit Test Related Information. We saw many students practicing unit tests at least once during the lab. During the feedback session, six students told us that they appreciated that code was provided based their own progress, which allowed them to quickly test if a new component was wired correctly. Performing a unit test gave students confidence that the completed part of the circuit was correct before they approached the rest of the circuit. It was interesting for us to observe that unit test was more practiced by experienced students (3/4). The students with less experience were not completely sure about the importance of unit test. Some thought it was for debugging and only performed a unit test when the instructor explicitly asked. It is thus important for the instructor to consider that instead of ambiently presenting the unit test suggestions, students should be notified explicitly to perform unit tests before proceeding in a circuit.

Overall, the students who had less of a background in electronics and circuits, were more attached to our system whenever there was a question. They used the system more proactively as a primary source of reference and they found the information easy to access. When the system was not in use, the students did not find the presence of the smartphone and the information provided to be distracting.

Instructor Experience

The instructor found that the way the component and module related help was provided to students made it easier for them to get a smoother start on their project. He said *"Clearly, I had far less students struggling at the beginning in comparison to the last lab. I did not change the way how I taught, so I think it was the context-sensitive information helped here."*

With less questions during the lab, the instructor had the opportunity to be proactive at looking at the students' progress and engaging with more students. The instructor told us that: *"I had almost no student asking me questions at the beginning of the lab. So I had the time to look around at their circuits and give suggestions when needed. I liked that I had the opportunity to engage with more students this way."*

Nevertheless, the instructor felt that it was still challenging to debug the students' circuits remotely. Most bugs appeared in the last twenty minutes of the lab, so students still had to wait to get help. While the instructor felt that some of the common circuit errors did not occur as many times as without the system, it was inevitable to have bugs completely out of his expectation (e.g. miswiring Arduino pin). As such, a debugging tool tailored to the remote classroom setting is still very much needed.

Finally, the instructor liked that circuit style behaviors could be reinforced in a more effective way. Traditionally, circuit styles are considered "non-essential" in teaching electronics labs because there are often other things (e.g., working principle of a component) considered to be more important. The instructor said: *"I found it hard to integrate circuit style behaviors into my teaching because it feels a bit out of the flow of my lecture. The system provides an alternative way, especially it allowed the reminder to be sent to the students based on their progress. I think it was effective."*

LIMITATION AND FUTURE WORK

This section presents the limitations of this work, lessons learned, and directions for future research.

Camera View

In our current implementation, the Arduino board was not placed inside the camera view by default, due to the height of the phone and viewing angle of the camera. This can be improved in the future using a detachable wide-angle lens. We foresee that the utility of the system can be further improved if the viewing angle of circuit can be adjustable by the student. Thus, our next step is to add a hinge to the 3D printed station to allow the smartphone to be tilted in different ways. This will likely improve the usability of our system by making it better for supporting the communication between students and the instructor for remote debugging scenarios. From the technical side, an adjustable camera view also introduces challenges in sensing different circuit components, which warrants future research.

Debugging

On a related note, the current implementation of our system does not provide a tool to support remote debugging for the instructor. This is the next step of our next research, as understandably, a tool that can automatically detect and inform the students about circuit errors can also be useful in remote labs. We plan to extend our machine learning algorithm to sense wires and the value of some of the common basic components (e.g. a resistor). Our preliminary study found that using Mask R-CNN, the value of resistors can be reliably detected by sensing the color strips of the resistor with an accuracy of 98.7%. This will eventually allow the system to replicate a student's circuit in a virtual world for error detection and reduction. Additionally, the functionality of the circuit can be verified using a software simulator before it is powered to prevent the components from being damaged.

Vision Based Sensing

In comparison to existing approaches [41], vision-based sensing does not need ad-hoc hardware and sensors. This makes it easier for the system to be adopted by a wide student population. The flip side is that sensing beyond components is less intuitive using vision. For example, our current implementation does not sense the orientation of the components or differentiate different models of the same component. To push it further, we plan to train another model using high-resolution images (e.g. 4k images) to capture more subtle details on the components, such as text labels. Additionally, we will explore 3D cameras as a means of enhancing the sensing capability of our system.

Incorporating AI

One perspective for our current system is from the angle of an intelligent agent that supports learning, practicing and teaching electronics labs in a remote context. While machine intelligence is currently used for understanding students' progress, more fruitful research can be carried out in applying AI and machine learning to better facilitate Q/A, error detection and reduction. On the other hand, it is also important to understand what aspects of a remote electronics lab can be better supported by a human or a machine and to understand how these two entities can work together to provide students with a better learning experience. We believe this line of research will lead to a much more effective hybrid system incorporating both machine and human intelligence to solve the pressing problems novices and others face in STEAM education and beyond.

System Evaluation

Our system evaluation was conducted with circuits replicated based on the Fritzing diagrams. We followed the same layout and location for the components described in the diagrams. Many of the circuits are well laid out on the breadboard in terms of component location and spacing. Future research will be conducted to validate the accuracy of our sensing techniques with components located at less common locations (e.g., corners) or clustered close to each other. This will give us insights into how well the system can be extended to support more features, such as sensing, detecting, and discouraging bad circuit styles.

CONCLUSION

In this work, we introduce the concept of context-sensitive remote learning resources for electronics labs, built on top of existing video tools. With context-sensitive information, our system enables both students and instructors in remote labs to manage class time, teaching and questions more effectively. To demonstrate the effectiveness of our CircuitLab system, we ran a remote study with one instructor and nine students with mixed levels of experience in electronics. We shared the insights gained from deploying our study and its design, and we believe that this work can inspire future research in the field of context-sensitive systems and tools not just for learning electronics, but also other areas of education and teaching.

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